Test of Mathematics for University Admission

DELUXE

Set 1 · Paper 2

20 questions · 75 minutes · no calculator

A paper by

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Twenty questions written as a prediction of the October 2026 Paper 2, built from the 2024, 2025 and 2026 papers rather than to a target difficulty. The question types, the option counts, the length of the stems and the spread of topics are taken from those papers. Full worked solutions and an answer key follow the paper.

Before you begin

If you have come here from Paper 1 you already know what this set is for. If you have not, sit Paper 1 first: the two are written to be taken as a pair, in one sitting, with a short break between them.

This is a prediction paper. It is not a collection of hard problems, and it is not trying to be harder than the real thing. Every question in it was written after reading the 2024, 2025 and 2026 papers and asking one narrow question: what is this examination actually going to put in front of you in October? So the proportion of necessary-and-sufficient items, the proportion of I/II/III items, the number of options, the length of the stems and the spread of topics are all taken from those papers rather than chosen for effect. Two or three questions here are harder than anything you will meet on the day. Two or three are easier. That is deliberate: a real paper is not a ramp.

Paper 2 is a different animal from Paper 1. Paper 1 asked whether you could find the short route through a calculation. This paper asks whether you can tell the difference between an argument and something that merely looks like one. Most questions hand you statements, conditions or complete arguments and ask you to decide exactly which of them survive scrutiny. The trap is almost never an arithmetic slip. It is a converse mistaken for the original, a quantifier read in the wrong order, a boundary case nobody checked, or a counterexample nobody bothered to look for.

Sit it properly: seventy-five minutes, no calculator, no notes, in one go. Then mark it against the table overleaf, which converts your raw mark to the TMUA's own scale. The number you get is meant to be an honest estimate of what you would score on the day. Treat it as an estimate, not as a verdict.

Then, for every question you missed or guessed, write the argument out yourself from a blank page. On this paper more than any other, "I would have got that" does not count. If you cannot write down the counterexample or the proof, you did not have it.

A word on method. When a statement looks true, try to break it before you try to prove it; when it looks false, try to prove it before you go hunting for a counterexample. Almost every question here punishes whichever of those two instincts you have by default.

One last thing. Consistency beats intensity. Two focused hours every day will take you further than a ten-hour Sunday followed by five days of nothing. The candidates who reach 9.0 are rarely the ones with the most talent. They are the ones who kept turning up after the bad mock, after the question that made no sense, after the week in which nothing seemed to improve. The score is built on the days you did not feel like it.

Keep showing up.

Viluxent (mkb6)

Candidate instructions

- You have 75 minutes to attempt all 20 questions.
- No calculator may be used. No formula sheet is provided or needed.
- Each question has exactly one correct answer among the options given. The number of options varies between questions, so read each list carefully before answering.
- Where statements are labelled I, II and III, choose the option that lists exactly the statements asked for.
- There is no penalty for a wrong answer, so attempt everything.

A note on style

This is a Paper 2. Every question is about reasoning: whether a statement must hold, whether a condition is necessary, sufficient, both or neither, whether an argument is valid, and where a proof first goes wrong. Between them the twenty questions draw on logic and quantifiers, induction and contradiction, polynomials, functions and their graphs, sequences, integration, coordinate geometry, trigonometry, probability, number, counting and invariants. Everything needed is on the specification; nothing is needed that is not.

The type mix follows the recent papers rather than a house style: eighteen reasoning items and two counting questions with a single numerical answer, because the real paper carries one or two of those as well. Option lists run from four to eight.

The options have been chosen adversarially. Where a question asks which statements are true, the false ones are statements that almost every candidate believes on first reading, and the true ones look too strong to be true. Testing a handful of values will usually confirm whatever you already believed, which is exactly why it does not work here.

Score calibration

Mark one point per question and read your raw mark across. The TMUA reports a score on a scale from 1.0 to 9.0, and in the middle of the range the conversion is close to linear.

17–20	scaled 8.0 to 9.0 — the top few per cent of the cohort
14–16	scaled 7.0 to 7.9 — clear of every published offer threshold
11–13	scaled 6.0 to 6.9 — the 6.5 borderline runs through this band
8–10	scaled 5.0 to 5.9 — the middle of the national cohort
0–7	below 5.0 — work through the solutions before you sit the next one

The conversion is indicative: the TMUA publishes its own raw-to-scaled table only after each sitting, and it moves by a few tenths from year to year with the difficulty of the paper and the strength of the cohort.

Question 1

Throughout, f denotes a polynomial with real coefficients.

Which of the following statements must be true?

I If the equation $f(x) = x$ has no real solution, then the equation $f\big(f(x)\big) = x$ has no real solution.

II If the equation $f(x) = x$ has exactly one real solution, then the equation $f\big(f(x)\big) = x$ has exactly one real solution.

III If the equation $f\big(f(x)\big) = x$ has exactly two distinct real solutions, then both of them satisfy $f(x) = x$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Let f be a function defined for every real number. Consider the two statements

P: for every real x there is a real y with $f(y) > f(x)$;

Q: there is no real number M such that $f(x) \leq M$ for every real x.

Which one of the following is correct?

A Q is necessary and sufficient for P.

B Q is necessary but not sufficient for P.

C Q is sufficient but not necessary for P.

D Q is neither necessary nor sufficient for P.

Question 3

For a positive integer n, let $S(n)$ denote the sum of the decimal digits of n. For example $S(144) = 9$. Which of the following statements are true?

I For every positive integer n that is divisible by 9, $S(n) = S(2n)$.

II For every positive integer n, if $S(n) = S(2n)$ then n is divisible by 9.

III There is no positive integer n with $S(n) = S(n+1)$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Question 4

For each positive integer n, $S(n)$ is a statement. Three sets of conditions are proposed, each intended to guarantee that $S(n)$ is true for every positive integer n.

I $S(1)$ is true; and for every $n \geq 1$, if $S(n)$ is true then at least one of $S(n+1)$ and $S(n+2)$ is true.

II $S(1)$ is true; for every $n \geq 1$, if $S(n)$ is true then $S(2n)$ is true; and for every $n \geq 2$, if $S(n)$ is true then $S(n-1)$ is true.

III $S(m)$ is true for at least one positive integer m; and for every $n \geq 1$, if $S(n+1)$ is true then $S(n)$ is true.

Which of these sets of conditions guarantee that $S(n)$ is true for every positive integer n?

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

In the statements below, A, B and C are events, each of which has probability strictly between 0 and 1, and B' and C' denote the complements of B and C. All the conditional probabilities written down are assumed to be defined.

Which of the following statements must be true?

I If $P(A \mid B) = P(A \mid B')$, then A and B are independent.

II If $P(A \mid B \cap C) > P(A \mid B' \cap C)$ and $P(A \mid B \cap C') > P(A \mid B' \cap C')$, then $P(A \mid B) > P(A \mid B')$.

III If $P(A \mid C) > P(A \mid C')$ and $P(B \mid C) > P(B \mid C')$, then $P(A \cap B \mid C) > P(A \cap B \mid C')$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

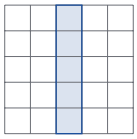
G II and III only

H I, II and III

Question 6

Each of the 25 squares of a 5×5 board holds a lamp, and every lamp is initially off. A *move* consists of choosing one row or one column and switching all five lamps in it: every lamp in that row or column that was off is turned on, and every lamp that was on is turned off.

How many of the 2^{25} possible configurations of the board can be reached by some sequence of moves?



one move

A 32

B 256

C 511

D 512

E 1024

F 2048

A sequence of real numbers $a_1, a_2, a_3, \ldots$ is called *subadditive* if

$$a_{m+n} \leq a_m + a_n \qquad \text{for all positive integers } m \text{ and } n.$$

Which of the following statements are true for every subadditive sequence?

I $a_n \leq n a_1$ for every $n \geq 1$.

II If $a_1 < 0$, then $a_n < 0$ for every $n \geq 1$.

III The sequence $\frac{a_1}{1}, \frac{a_2}{2}, \frac{a_3}{3}, \ldots$ is decreasing.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Question 8

Let f be a polynomial of degree 4 with real coefficients.

Which one of the following statements is true for every such f?

A If f has exactly two distinct real roots, then f' has exactly two distinct real roots.

B If f' has three distinct real roots, then f has at least two distinct real roots.

C If f'' has two distinct real roots, then f has at least one real root.

D If f has no real root, then f' has exactly one real root.

E If f' has exactly one real root, then f has exactly two distinct real roots.

F If f has four distinct real roots, then f'' has two distinct real roots.

Question 9

What is the smallest positive integer n with the following property?

> Every set of n different integers contains two different members whose sum is divisible by 100, or two different members whose difference is divisible by 100.

A 26

B 50

C 51

D 52

E 53

F 101

The function f is continuous and strictly increasing on $[0,1]$, with $f(0)=0$ and $f(1)=1$. Let g be the inverse function of f, so that g is also continuous and strictly increasing on $[0,1]$, with $g(0)=0$ and $g(1)=1$.

Which of the following statements must be true?

I $\displaystyle\int_0^1 f(x)\,dx + \int_0^1 g(x)\,dx = 1.$

II If $\displaystyle\int_0^1 f(x)\,dx = \tfrac{1}{2}$, then $f(x)=x$ for every $x \in [0,1]$.

III $\displaystyle\int_0^1 f(x)\,dx \leq \int_0^1 g(x)\,dx$ if and only if $f(x) \leq x$ for every $x \in [0,1]$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

The numbers $1, 2, 3, \ldots, 2026$ are written on a board. A *step* consists of choosing two of the numbers currently on the board, erasing them, and writing the non-negative difference of the two erased numbers in their place. After 2025 steps exactly one number remains.

Which of the following statements are true?

I Whatever choices are made, the number that remains is odd.

II Whatever choices are made, the number that remains is at most 2026.

III The choices can be made so that the number that remains is exactly 2025.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Consider the following claim.

> **Claim.** Let f be a polynomial with real coefficients such that $f(x) \geq 0$ for every real number x. Then there is a polynomial g with real coefficients such that $f(x) = \big(g(x)\big)^2$ for every real x.

A student offers the following argument.

(1) Either f is a non-negative constant, or f has even degree and positive leading coefficient; for otherwise $f(x)$ would be negative for x of large enough modulus.

(2) Every real root of f has even multiplicity, since f would otherwise change sign at that root.

(3) The remaining roots of f are not real, and they occur in complex conjugate pairs, because f has real coefficients.

(4) Hence $f(x) = k\big(\prod_i (x - r_i)^{m_i}\big)^2 \prod_j (x - \alpha_j)(x - \overline{\alpha_j})$, where $k > 0$, the r_i are the real roots and the α_j are the non-real ones.

(5) Each factor $(x - \alpha_j)(x - \overline{\alpha_j})$ is a real quadratic taking only positive values, so it equals $(x - p_j)^2 + q_j^2$ with $q_j \neq 0$, and is therefore the square of a polynomial with real coefficients.

(6) Multiplying these squares together and absorbing $\sqrt{k}$ gives $f = g^2$ with g real.

Which one of the following is true?

A The claim is true, and the argument is a correct proof of it.

B The claim is true, but the argument is not a correct proof; its first error is in line (2).

C The claim is false, and the first error in the argument is in line (1).

D The claim is false, and the first error in the argument is in line (2).

E The claim is false, and the first error in the argument is in line (4).

F The claim is false, and the first error in the argument is in line (5).

G The claim is false, and the first error in the argument is in line (6).

In the statements below, a, b and c denote positive integers, and $\gcd(a,b)$ denotes the greatest common divisor of a and b.

Which of the following statements are true?

I If a divides bc, then a divides b or a divides c.

II If $\gcd(a,b) = 1$ and a divides bc, then a divides c.

III If a divides c and b divides c, then ab divides $c \cdot \gcd(a,b)$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Question 14

Two players take turns removing stones from a single pile of n stones, and the player who removes the last stone wins. Both players play as well as possible, and the first player moves first.

In each of the statements below, the set shown is the set of numbers of stones that a player is allowed to remove in a single turn.

I With moves $\{1,2,4\}$, the first player has a winning strategy if and only if n is not a multiple of 3.

II With moves $\{1,2,3\}$, the first player has a winning strategy if and only if n is not a multiple of 4.

III With moves $\{1,3,4\}$, the first player has a winning strategy if and only if n is not a multiple of 7.

Which of these statements are true?

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Question 15

In a school there are exactly 30 students and a number of clubs. Every student belongs to exactly three of the clubs, and every club has exactly five members.

Which of the following statements must be true?

I There are exactly 18 clubs.

II Some two students belong to exactly the same three clubs.

III Some two clubs have no member in common.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Let a, b and c be positive real numbers, and consider the statement

$$*: \quad \text{there is a triangle whose side lengths are } a,\ b \text{ and } c.$$

Which one of the following conditions is necessary and sufficient for $*$?

A $a^2+b^2+c^2<2(ab+bc+ca)$

B $a^2+b^2+c^2>ab+bc+ca$

C $a^4+b^4+c^4<2\left(a^2b^2+b^2c^2+c^2a^2\right)$

D $a^2+b^2>c^2,\ b^2+c^2>a^2$ and $c^2+a^2>b^2$

E $ab+bc+ca>\dfrac{(a+b+c)^2}{4}$

F $(a+b+c)^3>27abc$

Question 17

In triangle ABC the angles at A, B and C are written A, B and C.

Which one of the following statements is true for every triangle ABC?

A $\cos A + \cos B + \cos C > 1$

B $\sin A + \sin B + \sin C > 2$

C $\cos^2 A + \cos^2 B + \cos^2 C \geq 1$

D $\sin A + \sin B + \sin C > \cos A + \cos B + \cos C$

E $\sin^2 A + \sin^2 B + \sin^2 C \leq 2$

F $\cos A \cos B \cos C > 0$

Question 18

Which one of the following conditions, on its own, guarantees that a triangle is equilateral?

A Two of its medians have the same length.

B Two of its altitudes have the same length.

C Two of its internal angle bisectors have the same length.

D Its centroid and its incentre are the same point.

E Its orthocentre lies inside it.

F It has a line of symmetry.

Claim. If n is a positive integer and $\sqrt{n}$ is not an integer, then $\sqrt{n}$ is irrational.

Three arguments are offered. Throughout, n is a positive integer whose square root is not an integer.

I Suppose $\sqrt{n} = p/q$ with p, q positive integers and $\gcd(p,q) = 1$. Then $nq^2 = p^2$, so q^2 divides p^2. But $\gcd(p,q) = 1$ forces $\gcd\left(p^2, q^2\right) = 1$, so $q^2 = 1$ and $\sqrt{n} = p$ is an integer, contrary to hypothesis.

II Suppose $\sqrt{n} = p/q$ with $\gcd(p,q) = 1$. Then $p^2 = nq^2$, so p^2 is a multiple of n, and hence p is a multiple of n. Writing $p = nr$ gives $n^2r^2 = nq^2$, so $q^2 = nr^2$ and q is a multiple of n too. This contradicts $\gcd(p,q) = 1$.

III Let $m = \lfloor \sqrt{n} \rfloor$, so that $m < \sqrt{n} < m+1$. Suppose $\sqrt{n}$ is rational and let q be the smallest positive integer for which $q\sqrt{n}$ is an integer. Put $q' = q\sqrt{n} - qm$. Then q' is an integer with $0 < q' < q$, and $q'\sqrt{n} = qn - mq\sqrt{n}$ is also an integer. This contradicts the minimality of q.

Which of the arguments are valid proofs of the claim?

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

For a positive integer n, let $R(n)$ denote the integer obtained by writing the decimal digits of n in the reverse order, any leading zeros then being discarded. For example $R(2178) = 8712$ and $R(1250) = 521$.

Which of the following statements are true?

I If n is divisible by 9, then $R(n)$ is divisible by 9.

II There is no positive integer n with $R(n) = 2n$.

III There is a positive integer n with $R(n) = 4n$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

Answer key

MARK BEFORE READING ON

Q	Ans	Value	Q	Ans	Value
1	**F**	I and III only	11	**H**	I, II and III
2	**C**	Q sufficient, not nec.	12	**F**	false; line (5)
3	**G**	II and III only	13	**G**	II and III only
4	**C**	II only	14	**E**	I and II only
5	**B**	I only	15	**F**	I and III only
6	**D**	512	16	**C**	quartic condition
7	**E**	I and II only	17	**A**	$\cos A + \cos B + \cos C > 1$
8	**F**	four roots $\Rightarrow f''$	18	**D**	centroid $=$ incentre
9	**D**	52	19	**F**	I and III only
10	**B**	I only	20	**H**	I, II and III

Mark honestly, then read the solution to every question you did not get out cleanly, including the ones you guessed correctly. On this paper a correct guess is worth nothing to you until you can write down the proof or the counterexample that justifies it.

Question 1 *answer F*

I is true. f is continuous, so $h(x) = f(x) - x$ is continuous. If $f(x) = x$ has no real solution then h is never zero, so by the intermediate value theorem h has constant sign. Say $h > 0$ throughout, i.e. $f(x) > x$ for every x. Then for any x,

$$f\big(f(x)\big) > f(x) > x,$$

where the first inequality applies the property at the point $f(x)$. So $f(f(x)) = x$ has no solution either. The case $h < 0$ is identical with the inequalities reversed. True.

III is true. Every solution of $f(x) = x$ is a solution of $f(f(x)) = x$. Suppose $f(f(x)) = x$ has exactly two distinct real solutions and that one of them, a, is not a fixed point. Put $b = f(a)$; then $b \neq a$ and $f(b) = a$, so $\{a, b\}$ is a genuine 2-cycle and b is also a solution of $f(f(x)) = x$. Hence the two solutions are exactly a and b, and neither is a fixed point. But $h(a) = b - a$ and $h(b) = a - b$ have opposite signs, so by the intermediate value theorem h vanishes somewhere strictly between a and b — giving a fixed point, which would be a third solution of $f(f(x)) = x$. Contradiction. True.

II is false. Take $f(x) = -x^3$. Then $f(x) = x$ reads $-x^3 = x$, i.e. $x\big(x^2 + 1\big) = 0$, whose only real solution is $x = 0$. But

$$f\big(f(x)\big) = -\big(-x^3\big)^3 = x^9,$$

so $f(f(x)) = x$ becomes $x^9 = x$, i.e. $x\big(x^8 - 1\big) = 0$, with three real solutions $x = 0, \pm 1$. False.

Why this is the trap. II is the converse of the true statement "$f(f(x)) = x$ has exactly one solution $\Rightarrow f(x) = x$ has exactly one solution", which does follow from I and III. Reading the implication in the comfortable direction is exactly the error the question is built around. Answer: I and III only.

Question 2 *answer C*

Q implies P. Q says f is not bounded above. Given any x, the number $f(x)$ is not an upper bound for the values of f, so there is a y with $f(y) > f(x)$. Hence Q is sufficient for P.

P does not imply Q. P says only that f attains no greatest value. A function can fail to attain a maximum while still being bounded above: take

$$f(x) = \frac{x^2}{x^2 + 1},$$

whose values fill $[0, 1)$. For every x there is a y with $f(y) > f(x)$, so P holds; but $f(x) < 1$ for all x, so f is bounded above and Q fails.

Conclusion. Q is sufficient but not necessary for P.

Remark. The distinction is between "no largest value" and "no upper bound" — a supremum that is never reached separates the two. Candidates who read P as "f increases without limit" answer A.

Question 3 *answer G*

The tool. $S(n) \equiv n \pmod 9$ for every n, because $10 \equiv 1 \pmod 9$.

II is true. If $S(n) = S(2n)$ then, reducing mod 9, $n \equiv 2n$, so $n \equiv 0 \pmod 9$. True.

III is true. If $S(n) = S(n+1)$ then $n \equiv n+1 \pmod 9$, i.e. $0 \equiv 1 \pmod 9$, which is absurd. So no such n exists. True.

I is false. It is the converse of II, and the congruence gives no information in that direction. A counterexample is $n = 144$: it is divisible by 9, and

$$S(144) = 9, \qquad S(288) = 2+8+8 = 18.$$

False. (The smallest counterexamples are $144, 234, 243, 324, \ldots$; every $n \le 143$ divisible by 9 does satisfy $S(n) = S(2n)$, which is precisely why the statement looks true.)

Remark. Doubling n multiplies the digit sum by 2 and then subtracts 9 for each carry, so $S(2n) = 2S(n) - 9c$ where c is the number of digits of n that are at least 5. Thus $S(n) = S(2n)$ exactly when $S(n) = 9c$, which is a genuine extra condition and not merely divisibility by 9. Answer: II and III only.

Question 4 *answer C*

II works. From $S(1)$ and the doubling rule, $S\left(2^k\right)$ is true for every k. Given any n, choose k with $2^k \ge n$; the descending rule then takes us from $S\left(2^k\right)$ down one step at a time to $S(n)$. So every $S(n)$ is true. Guarantees it.

I does not. The conclusion "at least one of $S(n+1)$, $S(n+2)$" never forces a particular statement. Let $S(n)$ be true exactly when $n = 1$ or n is even. Then $S(1)$ is true; and whenever $S(n)$ holds, one of $n+1$, $n+2$ is even, so one of $S(n+1)$, $S(n+2)$ holds. Yet $S(3)$ is false. Does not guarantee it.

III does not. The implication only runs downwards. Let $S(n)$ be true exactly when $n \le 5$. Then $S(3)$ is true, so the first condition holds; and if $S(n+1)$ is true then $n+1 \le 5$, so $n \le 5$ and $S(n)$ is true. Yet $S(6)$ is false. Does not guarantee it.

The moral. Induction needs a rule that reaches every integer. In II the doubling rule climbs arbitrarily high and the descending rule fills in everything below; in I nothing pins down a specific successor, and in III nothing ever goes up. Answer: II only.

Question 5 *answer B*

I is true. By the law of total probability,

$$P(A) = P(A \mid B)P(B) + P(A \mid B')P(B').$$

If the two conditional probabilities are equal, to p say, the right-hand side is $p\big(P(B) + P(B')\big) = p$. So $P(A) = p = P(A \mid B)$, which rearranges to $P(A \cap B) = P(A)P(B)$. True.

II is false: this is Simpson's paradox. An advantage inside every stratum can be reversed overall, if the two groups are unequally distributed between the strata. Take

	C (successes/trials)	C' (successes/trials)	combined
B	$81/87 \approx 0.93$	$192/263 \approx 0.73$	$273/350 = 0.780$
B'	$234/270 \approx 0.87$	$55/80 \approx 0.69$	$289/350 \approx 0.826$

Here B beats B' within C and within C', yet loses overall, because B is concentrated in the stratum C' where success is rare. All eight cell probabilities are strictly between 0 and 1. False.

III is false. Conditioning on C can raise the chance of A and the chance of B separately while making them compete, so that they rarely occur together. Concretely, let $P(C) = \frac{1}{2}$ and, given C, let A and B each have probability $\frac{1}{2}$ and never occur together; given C', let A and B each have probability $\frac{2}{5}$ and always occur together. Then

$$P(A \mid C) = \tfrac{1}{2} > \tfrac{2}{5} = P(A \mid C'), \qquad P(B \mid C) = \tfrac{1}{2} > \tfrac{2}{5} = P(B \mid C'),$$

while $P(A \cap B \mid C) = 0 < \frac{2}{5} = P(A \cap B \mid C')$. Every event here has probability strictly between 0 and 1: $P(A) = P(B) = 0.45$ and $P(A \cap B) = 0.2$. False.

Answer. I only. Statement I looks like the weakest of the three and is the only one that survives.

Question 6 *answer D*

Only the parity of the counts matters. Switching a row twice restores it, and the moves commute. So the outcome of any sequence of moves is determined by the set R of rows switched an odd number of times and the set C of columns switched an odd number of times. Lamp (i, j) ends up on exactly when $i \in R$ and $j \notin C$, or $i \notin R$ and $j \in C$ — that is, when exactly one of $i \in R$, $j \in C$ holds.

Count the pairs, then remove the double count. There are $2^5 \cdot 2^5 = 1024$ pairs (R, C). Two pairs give the same configuration precisely when one is obtained from the other by complementing both sets: replacing (R, C) by (R^c, C^c) leaves "exactly one of the two memberships holds" unchanged, and no other coincidence occurs. So the pairs fall into classes of size exactly 2, and the number of reachable configurations is

$$\frac{1024}{2} = 512.$$

Traps. 1024 forgets the double count; 32 counts only row moves; 511 wrongly excludes the all-off board, which is of course reachable by doing nothing.

Question 7 *answer E*

I is true. Induct on n. It holds at $n = 1$, and if $a_n \le na_1$ then $a_{n+1} \le a_n + a_1 \le (n+1)a_1$. True.

II is true. Apply subadditivity with $m = 1$: $a_{n+1} \le a_n + a_1 < a_n$, so the sequence is strictly decreasing; and $a_1 < 0$, so every term is at most $a_1 < 0$. (Equally, I gives $a_n \le na_1 < 0$ directly.) True.

III is false. Define

$$a_n = \begin{cases} 1, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

This is subadditive: if m and n have the same parity then $m+n$ is even and $a_{m+n} = 0 \le a_m + a_n$; if they have opposite parity then $m+n$ is odd and $a_{m+n} = 1 = 1+0 = a_m + a_n$. But

$$\frac{a_1}{1} = 1, \quad \frac{a_2}{2} = 0, \quad \frac{a_3}{3} = \tfrac{1}{3}, \quad \frac{a_4}{4} = 0, \quad \frac{a_5}{5} = \tfrac{1}{5}, \ \ldots$$

is not decreasing, since it rises from 0 to $\frac{1}{3}$. False.

Remark. What is true — and what makes III so tempting — is that a_n/n always converges, to its own infimum. Convergence is not monotonicity, and here the even and odd terms interleave. Answer: I and II only.

Question 8 *answer F*

F is true. If f has four distinct real roots, Rolle's theorem gives a root of f' strictly inside each of the three gaps, so the cubic f' has three distinct real roots. Applying Rolle again to f' gives a root of f'' strictly inside each of its two gaps, so the quadratic f'' has two distinct real roots. True.

Every other option fails.

- A. $f(x) = \left(x^2 - 1\right)^2$ has exactly two distinct real roots, but $f'(x) = 4x\left(x^2 - 1\right)$ has three.
- B. $f(x) = x^4 - 2x^2 + 10$ has $f'(x) = 4x^3 - 4x$ with three distinct real roots, yet $f(x) = \left(x^2 - 1\right)^2 + 9 > 0$ has none.
- C and D. $f(x) = x^4 - x^2 + 1$ has no real root, since $u^2 - u + 1 > 0$ for all real $u = x^2$. But $f''(x) = 12x^2 - 2$ has two distinct real roots, killing C, and $f'(x) = 4x^3 - 2x$ has three distinct real roots, killing D.
- E. $f(x) = x^4 + 1$ has $f'(x) = 4x^3$, whose only real root is 0, yet f has no real root at all.

Trap. E is true with "at most" in place of "exactly": one turning point does force at most two roots. Strengthening it to "exactly" is the error.

Question 9 *answer D*

Work modulo 100. Two integers have a difference divisible by 100 exactly when they are congruent mod 100; they have a sum divisible by 100 exactly when their residues r and s satisfy $r+s \equiv 0$, i.e. $s \equiv -r$.

Group the residues. Partition $\{0,1,\ldots,99\}$ into the classes $\{r, 100-r\}$:

$$\{0\}, \qquad \{50\}, \qquad \{1,99\},\ \{2,98\},\ \ldots,\ \{49,51\}.$$

That is $2+49=51$ classes.

The extremal set. A set avoiding both conditions can contain at most one member from each class. Indeed, two members in the same class either share a residue (difference divisible by 100) or have residues r and $100-r$ (sum divisible by 100); the singleton classes $\{0\}$ and $\{50\}$ fail on both counts at once. So such a set has at most 51 elements, and 51 is achieved, for example by $\{0,50,1,2,\ldots,49\}$.

Conclusion. Every set of 52 integers must contain such a pair, and 51 need not, so the smallest n is 52.

Traps. 51 is the size of the largest bad set — the answer to the question one step earlier. 50 and 26 come from forgetting the two self-paired residues 0 and 50.

Question 10 *answer B*

I is true. The graph of g is the graph of f reflected in the line $y=x$, so $\int_0^1 g$ is the area of the region between the graph of f and the y-axis, while $\int_0^1 f$ is the area between the graph and the x-axis. Because f runs from $(0,0)$ to $(1,1)$ and is strictly increasing, these two regions are disjoint and together fill the unit square:

$$\int_0^1 f(x)\,dx + \int_0^1 g(x)\,dx = 1.$$

True.

II is false. Having the right average does not pin down the function. Take

$$f(x) = x + \frac{\sin 2\pi x}{4\pi},$$

which satisfies $f(0)=0$, $f(1)=1$ and $f'(x) = 1+\frac{1}{2}\cos 2\pi x \geq \frac{1}{2} > 0$, so it is strictly increasing. Since $\int_0^1 \sin 2\pi x\,dx = 0$, we get $\int_0^1 f = \frac{1}{2}$, yet $f\left(\frac{1}{4}\right) = \frac{1}{4}+\frac{1}{4\pi} \neq \frac{1}{4}$. False.

III is false. The "if" direction is fine, but the "only if" fails, and the same f shows it: there $\int_0^1 f = \frac{1}{2} = \int_0^1 g$ by I, so the inequality between the integrals holds, while $f(x) > x$ for $0 < x < \frac{1}{2}$. An integral inequality is an average statement and cannot force a pointwise one. False.

Answer. I only.

Question 11 *answer H*

I: a parity invariant. Replacing a and b by $|a-b|$ changes the total on the board by

$$a+b-|a-b| = 2\min(a,b),$$

which is even. So the parity of the total is never altered. Initially the total is

$$1+2+\cdots+2026 = \frac{2026\cdot 2027}{2} = 1013\cdot 2027,$$

a product of two odd numbers and therefore odd. The single surviving number has the same parity, so it is odd. True.

II: the maximum never rises. Since $|a-b| \le \max(a,b)$, no step ever introduces a number larger than the largest already present. The largest starting number is 2026, so the final number is at most 2026. True.

III: a construction. Group the 2024 consecutive integers $2,3,\ldots,2025$ into 506 blocks of four consecutive numbers. In each block $\{k,k+1,k+2,k+3\}$ take $|k-(k+1)| = 1$ and $|(k+2)-(k+3)| = 1$, then $|1-1| = 0$. This reduces the whole range $2,\ldots,2025$ to a collection of zeros, which collapse to a single 0. The board now holds 1, 2026 and 0. Finally

$$|2026-1| = 2025, \qquad |2025-0| = 2025.$$

So 2025 is attainable, and it is odd and at most 2026, as I and II require. True.

Answer. I, II and III. The parity argument makes I and II easy to believe and III easy to doubt; the block construction is the only part that needs building.

Question 12 *answer F*

The claim is false. Take $f(x) = x^2+1$. It is non-negative everywhere, but it is not the square of a real polynomial: $f = g^2$ would force $\deg g = 1$, say $g = \alpha x+\beta$, giving $\alpha^2x^2+2\alpha\beta x+\beta^2 = x^2+1$ — so $\alpha = \pm 1$, $\beta = \pm 1$ and $2\alpha\beta = 0$, which is impossible.

Lines (1)–(4) are correct. A polynomial that is non-negative on the whole real line cannot have odd degree, and its leading coefficient must be positive; a real root of odd multiplicity is a sign change; non-real roots of a real polynomial come in conjugate pairs; and collecting these gives exactly the factorisation in line (4).

Line (5) is the error. The quadratic $(x-p_j)^2+q_j^2$ with $q_j \neq 0$ is indeed a real quadratic taking only positive values — but it is not the square of a real polynomial. A real square of degree 2 is $(\alpha x+\beta)^2$, which has a real root; a positive quadratic has none. The step confuses "is everywhere positive" with "is a square".

What is true instead. Each such factor is a sum of two squares, and the correct theorem is that a non-negative real polynomial is a sum of two squares of real polynomials — which is why $x^2+1 = x^2+1^2$ resists being written as one. Answer: the claim is false, and the first error is in line (5).

Question 13 *answer G*

I is false. Divisibility does not distribute over a product without a coprimality hypothesis. Take $a = 4$, $b = 2$, $c = 6$: then $a \mid bc$ since $4 \mid 12$, but $4 \nmid 2$ and $4 \nmid 6$. False.

II is true. This is Euclid's lemma in its general form. Since $\gcd(a,b) = 1$, there are integers u, v with $au + bv = 1$. Multiplying by c gives $c = acu + (bc)v$. Now a divides acu, and a divides bc by hypothesis, hence a divides $(bc)v$; so a divides the sum, which is c. True.

III is true. If $a \mid c$ and $b \mid c$ then $\operatorname{lcm}(a,b) \mid c$, so we may write $c = \operatorname{lcm}(a,b) \cdot k$ for some positive integer k. Using $\operatorname{lcm}(a,b) \cdot \gcd(a,b) = ab$,

$$c \cdot \gcd(a,b) = \operatorname{lcm}(a,b)\gcd(a,b) \cdot k = abk,$$

which is a multiple of ab. True.

Sanity check on III. With $a = 4$, $b = 6$, $c = 12$: $\gcd = 2$, and $c\gcd = 24 = ab$, so $ab \mid c\gcd$ with quotient 1. The statement is sharp, not merely a weak bound. Answer: II and III only.

Question 14 *answer E*

The method. Call a position *losing* if the player about to move loses under best play. A position is losing exactly when every legal move leads to a winning position, and 0 is losing.

I: moves $\{1,2,4\}$. Working upwards, 0 is losing; 1, 2 are winning (take the lot); 3 is losing, since $3-1=2$ and $3-2=1$ are winning and 4 is not a legal move; 4, 5 are winning; 6 is losing, since $6-1=5$, $6-2=4$ and $6-4=2$ are all winning. The pattern repeats with period 3: the losing positions are exactly the multiples of 3. So the first player wins precisely when $3 \nmid n$. True.

II: moves $\{1,2,3\}$. From any n the mover can leave any of $n-1, n-2, n-3$, so a player facing a multiple of 4 must leave a non-multiple, and a player facing a non-multiple can always return to a multiple of 4. The losing positions are the multiples of 4. True.

III: moves $\{1,3,4\}$ — and this is where it breaks. Computing as before:

n	0	1	2	3	4	5	6	7	8	9	10
position	L	W	L	W	W	W	W	L	W	L	W

The losing positions are $0, 2, 7, 9, 14, 16, \ldots$: those n congruent to 0 or 2 modulo 7. So the first player loses at $n = 2$, which is not a multiple of 7, and the stated criterion is wrong. False.

Remark. The neat answer for $\{1,2,\ldots,k\}$ — losing exactly at multiples of $k+1$ — tempts one to expect a single residue class whenever the move set has k elements. A move set with a gap in it generally produces a longer period and more than one losing class, as here. Answer: I and II only.

Question 15 *answer F*

I: count the memberships twice. Each of the 30 students contributes 3 memberships, giving 90 in total; each club accounts for exactly 5 of them. Hence the number of clubs is $90/5 = 18$. True.

III: count the intersecting pairs. Each student lies in $\binom{3}{2} = 3$ pairs of clubs, so at most $30 \times 3 = 90$ pairs of clubs can share a member. But there are $\binom{18}{2} = 153$ pairs of clubs altogether, and $153 > 90$. So at least 63 pairs of clubs share nobody; in particular some two clubs have no member in common. True.

II is false, and here is an arrangement that shows it. Label the 18 clubs by pairs (r,c) with $r \in \{1,2,3\}$ and $c \in \{1,2,\dots,6\}$, and give each student one club from each row r. Number the students $i = 0,1,\dots,29$ and write $i = 6j+t$ with $0 \le t \le 5$ and $0 \le j \le 4$. Send student i to the three clubs

$$(1,t), \qquad (2,t+j), \qquad (3,t+2j),$$

with the second coordinates read modulo 6. Within each block of six students t runs over all six values, so each club receives exactly one student per block and hence 5 in total, as required. Two students in the same block differ in t and so differ in their first club; two students in different blocks $j \neq j'$ with the same t differ in their second club, since $j \not\equiv j' \pmod 6$. So all 30 triples are distinct. False.

Answer. I and III only. Note that II and III pull in opposite directions and both look like pigeonhole statements; only one of them actually is.

Question 16 *answer C*

The identity behind C. For all real a,b,c,

$$2\left(a^2b^2+b^2c^2+c^2a^2\right) - \left(a^4+b^4+c^4\right) = (a+b+c)(-a+b+c)(a-b+c)(a+b-c).$$

With $a,b,c > 0$ the first bracket is positive, so condition C says exactly that

$$(-a+b+c)(a-b+c)(a+b-c) > 0.$$

Now two of these three brackets cannot both be negative: adding any two of them gives $2a$, $2b$ or $2c$, which is positive. So the product is positive precisely when all three are positive — which is the triangle inequality. Hence C is necessary and sufficient. (It is Heron's formula in disguise: the left-hand side is 16 times the square of the area.)

Why the others fail.

- A is necessary but not sufficient: with $(a,b,c) = (1,1,3)$ we get $1+1+9 = 11 < 14 = 2(1+3+3)$, yet $1+1 < 3$.
- B holds for all positive a,b,c unless $a = b = c$, so it says almost nothing.
- D is sufficient but not necessary: it characterises acute triangles, and excludes the right-angled $(3,4,5)$.
- E fails on the same $(1,1,3)$: $ab+bc+ca = 7 > \frac{25}{4}$, but there is no triangle.
- F is just AM–GM, true whenever a,b,c are not all equal.

Question 17 *answer A*

A is true. Using $C = \pi - A - B$ and the factor formulae,

$$\cos A + \cos B + \cos C = 1 + 4 \sin \tfrac{A}{2} \sin \tfrac{B}{2} \sin \tfrac{C}{2},$$

and each half-angle lies strictly between 0 and $\frac{\pi}{2}$, so each sine is strictly positive. Hence the sum exceeds 1 for every triangle. (Equivalently, the sum is $1 + r/R > 1$.) True.

The others all fail.

- B and D fail on a very "thin" triangle. Let A and B tend to 0, so $C \to \pi$. Then $\sin A + \sin B + \sin C \to 0$, which is neither greater than 2 nor greater than $\cos A + \cos B + \cos C \to 1$.
- C fails for the equilateral triangle: $3\cos^2 60^\circ = \frac{3}{4} < 1$. (The identity $\cos^2 A + \cos^2 B + \cos^2 C + 2\cos A \cos B \cos C = 1$ shows the expression exceeds 1 exactly for obtuse triangles.)
- E fails for the equilateral triangle too: $3\sin^2 60^\circ = \frac{9}{4} > 2$.
- F fails for any obtuse triangle, where exactly one cosine is negative.

Remark. Both B and E are true for a restricted family — B for triangles that are not too thin, E for non-acute ones — which is exactly why testing one or two familiar triangles is worthless here.

Question 18 *answer D*

D works. Place the vertices at position vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$. The centroid is $\frac{1}{3}(\mathbf{a} + \mathbf{b} + \mathbf{c})$ and the incentre is $\dfrac{a\mathbf{a} + b\mathbf{b} + c\mathbf{c}}{a + b + c}$, where a, b, c are the side lengths. These agree exactly when the weights are proportional, i.e. when $a = b = c$. So the triangle is equilateral. Guarantees it.

A synthetic route: the median from A and the bisector from A both pass through the common point, and they meet BC at the same point only if that point is both the midpoint and the point dividing BC in the ratio $AB : AC$ — forcing $AB = AC$. Applying this at two vertices gives all three sides equal.

Everything else gives only isosceles, or less.

- A: two equal medians force the two corresponding sides equal, so the triangle is isosceles — for example $(5, 5, 6)$ has two equal medians and is not equilateral.
- B: two equal altitudes force two equal sides, since $2 \times \text{area} = ah_a = bh_b$. Again only isosceles.
- C: two equal internal bisectors force two equal sides — this is the Steiner–Lehmus theorem — and no more.
- E: the orthocentre lies inside exactly when the triangle is acute, which is far weaker.
- F: a line of symmetry is precisely the isosceles condition.

Trap. Each of A, B, C and F is a genuine theorem whose conclusion is "isosceles". Reading the conclusion as "equilateral" — the commonest error on this kind of question — makes four options look correct at once.

Question 19 *answer F*

I is valid. If $\gcd(p,q)=1$ then p^2 and q^2 share no prime factor, so $\gcd(p^2,q^2)=1$. From $nq^2=p^2$ we get $q^2 \mid p^2$, and a divisor of p^2 that is coprime to p^2 must be 1. Hence $q=1$ and $\sqrt{n}=p$ is an integer, contradicting the hypothesis. Every step is justified. Valid.

III is valid. This is the minimal-denominator argument. Since $m<\sqrt{n}<m+1$, we have $0<\sqrt{n}-m<1$, so $q'=q\sqrt{n}-qm$ satisfies $0<q'<q$, and it is an integer because $q\sqrt{n}$ and qm are. Moreover

$$q'\sqrt{n}=qn-m\left(q\sqrt{n}\right)$$

is an integer as well. So q' is a smaller positive integer for which $q'\sqrt{n}$ is an integer, contradicting minimality. Valid.

II is not valid. The step "p^2 is a multiple of n, hence p is a multiple of n" is false in general. It holds when n is prime, but not otherwise: with $n=8$, $p=4$ we have $n \mid p^2$ (as $8 \mid 16$) while $n \nmid p$. Since the claim concerns all non-square n, an argument that silently assumes n prime proves nothing.

That the conclusion happens to be true does not rescue the argument: a proof must be valid for every n covered by the statement. Not valid.

Answer. I and III only. The familiar $\sqrt{2}$ proof is argument II with $n=2$, where the faulty step is legitimate — which is exactly why it is so easy to carry across without noticing.

Question 20 *answer H*

I is true. Reversing the digits does not change the multiset of digits, so it does not change the digit sum; discarding leading zeros does not change it either. Since $n \equiv S(n) \pmod 9$ and $R(n) \equiv S(n) \pmod 9$, we get $R(n) \equiv n \pmod 9$. So $9 \mid n$ gives $9 \mid R(n)$. True.

II is true. Suppose $R(n)=2n$, and let n have k digits $d_1 d_2 \ldots d_k$ with $d_1 \neq 0$.

If $d_k=0$ then $R(n)$ has fewer than k digits, so $R(n)<n<2n$. Hence $d_k \neq 0$ and $R(n)$ also has k digits. Since $2n$ has k digits, $d_1 \leq 4$; and the last digit of $2n$ is even while the last digit of $R(n)$ is d_1, so d_1 is even, giving $d_1 \in \{2,4\}$. As $n \geq d_1 \cdot 10^{k-1}$, we have $2n \geq 2d_1 \cdot 10^{k-1}$, so the leading digit of $2n$ — which must equal d_k — satisfies $d_k \geq 2d_1$. Also the final digit of $2n$ is the last digit of $2d_k$, and it must equal d_1.

- If $d_1=4$ then $d_k \geq 8$, and $2d_k$ must end in 4, forcing $d_k \in \{2,7\}$ — impossible.
- If $d_1=2$ then $d_k \geq 4$, and $2d_k$ must end in 2, forcing $d_k \in \{1,6\}$, so $d_k=6$. But then $2 \cdot 10^{k-1} \leq n < 3 \cdot 10^{k-1}$, so $4 \cdot 10^{k-1} \leq 2n < 6 \cdot 10^{k-1}$ and the leading digit of $2n$ is 4 or 5, not 6. Impossible.

So no such n exists. True.

III is true. Despite II, the analogous equation with 4 does have solutions:

$$R(2178)=8712=4\times 2178,$$

and 21978, 219978, 2199978, … all work as well. True.

Answer. I, II and III. The trap is to prove II, decide that the same argument "ought" to rule out 4, and answer I and II only. The leading-digit squeeze that kills the factor 2 leaves just enough room at the factor 4.

Provenance and syllabus notes

This set is a prediction, not an anthology. Every question was written after reading the 2024, 2025 and 2026 papers, and the shape of the paper is copied from them rather than chosen. Three features of those papers drive almost all of it. The first is the necessary-and-sufficient item, which asks for one of four verdicts and is the single most characteristic question type of this paper; Question 2 is built on it, Question 16 asks the same thing in words, and Questions 8, 17 and 18 are the one-sided variant in which six candidate implications are offered and exactly one survives. The second is the I/II/III lattice with an eight-option answer list, which those papers use for roughly one question in three. The third is the flawed-argument item, in which a complete proof is printed with numbered lines and the candidate must say both whether the claim is true and where the argument first fails: Question 12 is that question, and Question 19 is its close relative.

On syllabus

Every question sits inside the TMUA Paper 2 specification. The following were checked explicitly.

- No limits and no continuity arguments are examined beyond the intermediate value theorem, used in Questions 1 and 10. Question 7 uses only the definition of a bounded sequence, and that definition is supplied in the question.
- No modular arithmetic is assumed. Questions 3, 9 and 20 are phrased with digit sums, remainders and divisibility; the congruences in their solutions are shorthand for arguments with remainders.
- Probability is limited to conditional probability and the law of total probability (Question 5). Question 10 needs only the reflection of a graph in the line $y = x$, and Question 8 only the fact that between two roots of a polynomial there is a stationary point.
- No named result is needed anywhere. Euclid's lemma, Steiner–Lehmus and Simpson's paradox are named in the solutions for provenance only; every solution is self-contained.

On the answer options

Every option list was built after the solution, from the wrong answers the question actually produces. In the statement questions, each false statement is one that a strong candidate believes on first reading: a converse (Questions 1 and 3), an average confused with a pointwise bound (Question 10), a stratified comparison assumed to survive aggregation (Question 5), a monotonicity read into a convergence (Question 7), or a pattern that holds for one move set and is assumed to hold for the next (Question 14). Each true statement is one that looks too strong to hold. Question 11 deliberately has "all" as its answer, and the numerical questions carry the off-by-one answer next to the right one: 51 sits beside 52 in Question 9, and 1024 and 511 sit beside 512 in Question 6.

Viluxent mkb6

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